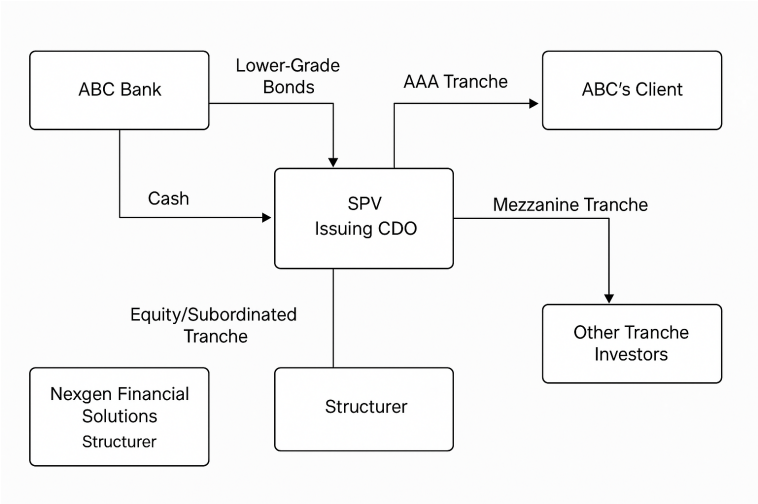
Risk Management Assignment 3

Due: 03. 12. 2025

Question 1

Prepare two diagrams. One should show all parties involved and all transfers of assets (including cash) among the various parties; specify amounts as far as known. The other diagram should represent the balance sheet of the SPV issuing the CDO, and show the various assets and liabilities.



Slide 2: SPV Balance Sheet

ASSETS (10,000 TCHF)	LIABILITIES (10,000 TCHF)
Bond Portfolio	1. Senior Tranche (AAA)
- 100 Lower-Grade Bonds	~9,800 TCHF (98%)
- Diversified Industries	
- Total: 10,000 TCHF	2. Mezzanine Tranche(s)
	~100 TCHF (1%)
	3. Equity Tranche
	~100 TCHF (1%) - First Loss
Total Assets: 10,000	Total Liabilities: 10,000

Question 2

Determine the expected loss of the portfolio. Who should, in your view, bear the economic impact of expected losses? Explain.

For the expected loss of each bond we use the following formula:

$$EL_i = \text{Exposure}_i \times \text{Probability of Default}_i \times (1 - \text{Recovery Rate}_i)$$

We assume we have 100'000 CHF invested in each bond. Thus, our exposure is 100'000 CHF. The recovery rate is available in the provided data, as well as the default probability. Summing up all individual expected losses gives us the total expected loss of 1'398.37 CHF.

The economic impact of the Expected Loss should be borne by the Equity Tranche investors, i.e. Nexgen. CDO tranches are structured by credit priority and the Equity Tranche is the most junior and absorbs the first losses up to its full size. The EL represents the statistically average loss for the portfolio. By structuring the Equity Tranche to be at least equal to the portfolio's EL, investors in all Mezzanine and Senior tranches are protected against the most probable losses, allowing them to receive higher credit ratings.

Question 3

Determine the Credit-Value-at-Risk (CVaR) for the default risk in the portfolio using a 99% and a 99.9% confidence level. Specify your assumptions about correlations/diversification effects, as well as about any other parameters used.

With a total exposure of 10'000'000, the $VaR_{99} = 699'587$ and the $VaR_{99.9} = 899'645$. This was computed using the credit Suisse risk model.

We had the following assumptions: Exposure for each bond is 100'000 CHF. For the standard deviations we used the heuristic from the risk model, namely multiplying the probability of default by 0.5. We divided the sectors in 7 groups by approximating the NOGA sectors of the Swiss Federation.

Question 4

Knowing that S&P's ratings reflect the "likelihood of payment – capacity [...] to meet the financial commitment on an obligation in accordance with the terms of the obligation", what conditions should a CDO tranche in your view satisfy in order to get a AAA-rating?

S&P's AAA rating signifies the highest likelihood of payment and an extremely strong capacity to meet the financial commitments of the obligation.A CDO tranche should satisfy the following conditions to attain a AAA-rating:

- Sufficient Credit Enhancement (Protection): The AAA tranche must be protected by a thick layer of lower-rated tranches (Equity and Mezzanine) that absorb nearly all plausible losses, including the CVaR at a very high confidence level (e.g., 99.9% or higher).The total size of the junior tranches (the attachment point of the AAA tranche) must be greater than the maximum loss expected in a severe economic downturn (the $CVaR_{99.9\%}$)

- Stress-Case Resilience: The tranche must demonstrate minimal loss even under extreme stress scenarios, such as the simultaneous default of a significant number of counterparties, as defined by the rating agency's models (e.g., 50% or more of the portfolio in a very high-stress simulation)

- Strong Portfolio Diversification: The underlying collateral should be highly diversified across different industries and geographies to minimize the impact of systematic risk factors. Our portfolio has diversification across 100 names and multiple industries.

- Legal/Structural Integrity: The SPV structure must be bankruptcy-remote, legally robust, and transparent to ensure the cash flows from the underlying assets are ring-fenced and directed to the AAA tranche holders with absolute priority

Question 5

Propose a CDO structure (including the specification of attachment and detachment points for all tranches) that allows the most senior tranche to get a AAA-rating, The senior tranche should be as large as reasonably possible, to get an attractive deal for the client. Explain your calculations.

To achieve a AAA rating for the senior tranche, its attachment point (the total size of all junior tranches) must be set high enough to cover the Credit-Value-at-Risk at a conservative level (e.g., 99.9%)

A plausible structure consistent with the CR+ loss distribution and the AAA default threshold is: Equity 0–3%, Mezz 1 3–6%, Mezz 2 6–10%, and a large AAA senior tranche 10–100%. The 10% subordination is chosen so that the model-implied probability of portfolio loss exceeding 10% is below the 0.3% 5-year AAA default rate.

Question 6

Quantify the level of coupons you propose to pay to the investors of the various CDO tranches. For what services delivered, costs incurred, risks taken, ... should Nexgen be remunerated? What cash flows / asset flows do you propose to exchange at inception of the transaction to have a "fair" deal? Explain.

Assign small AAA spreads (≈10–15 bp), higher mezz spreads (≈80–120 bp and 200–300 bp), and let Nexgen earn a high expected IRR via the equity tranche and a modest structuring fee. Nexgen is being remunerated for first-loss risk and for providing the structuring, modelling, and monitoring services. The initial flows are fair if the client's new AAA position has a much lower loss probability than the original bond portfolio, the mezz investors receive compensation consistent with the CR+ tail loss results, and the equity's expected return covers Nexgen's capital at risk and effort.

Appendix